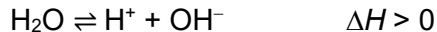


18 Water dissociates as shown.



At 25 °C, the concentration of water is given as 55.6 mol dm⁻³.

Which of the following statements are correct?

- 1 At 25 °C, the $\text{p}K_{\text{a}}$ of water is 14.
- 2 At 25 °C, the value of $\text{pH} < \text{p}K_{\text{w}} < \text{p}K_{\text{a}}$ of water.
- 3 At 50 °C, the value of $\text{p}K_{\text{w}}$ is lower than 14.

- A** 1 and 2 only
- B** 1 and 3 only
- C** 2 and 3 only
- D** 3 only